

SHI FENG

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Education and Qualification

Technical University of Munich

Postdoctoral Fellow in Quantum Matter and Quantum Information Theory Group

◦ Advisors: M. Knap, J. Knolle and F. Pollmann

Garching, Germany

2024–Present

The Ohio State University

Ph.D in Theoretical Condensed Matter Physics

◦ Advisor: N. Trivedi

◦ [Dissertation](#): Fractionalization in Frustrated Quantum Matter

Columbus, Ohio, USA

2018–2024

Xi'an Jiaotong University

B.S. in Physics (Honors Program), Qian Xuesen College

◦ Visiting student in UCR and UCLA, 2016–2018

Xi'an, Shaanxi, China

2014–2018

Interest and Expertise

1. Theoretical study of quantum spin liquids and topologically ordered matter
2. Non-equilibrium quantum dynamics, non-linear response and their application in condensed matter
3. Anderson localization, many-body localization, super-/sub-diffusion and anomalous transport
4. Quantum magnetism and frustrated systems: phase transitions, spin and energy transport
5. Tensor network methods for many-body systems: MPS, DMRG, time-evolution of tensor-network states
6. Statistical models and machine learning methods relevant for condensed matter theory

Publications and Preprints

1. *Magnetic field induced phenomena in Kitaev spin liquids*
S. Feng, N. Trivedi
[arXiv:2601.14496](#) (2026)
2. *Triangular J_1 - J_2 Heisenberg antiferromagnet in a magnetic field*
T. Bader, S. Feng, S. Budaraju, B. Federico, J. Knolle, F. Pollmann
[arXiv:2512.08116](#) (2025)
3. *FFT-accelerated auxiliary variable MCMC for fermionic lattice models*
D. Kong, S. Feng, J. Xie, Y. N. Wu
[arXiv:2510.13866](#) (2025)
4. *Transient localization from fractionalization: vanishingly small conductivity in gapless quantum magnets*
S. Feng, P. Zhu, J. Knolle, M. Knap
[arXiv:2509.07062](#) (2025)
5. *Finite spinon density-of-states in triangular-lattice delafossite TlYbSe_2*
B. Belbase, A. Unnikrishnan, S. Feng, E. S. Choi, J. Knolle, A. Banerjee
[arXiv:2504.05436](#) (2025)
6. *Obstruction to broken symmetries in topological flat bands*
P. Zhu, S. Feng, Y.-M. Lu
[Phys. Rev. B 111, L161411](#) (2025) [[arxiv:2408.14533](#)]
7. *Emergent quantum Majorana metal from a chiral spin liquid*
P. Zhu[†], S. Feng[†], K. Wang[†], T. Xiang, N. Trivedi ([†]Contributed Equally)
[Nat. Commun. 16, 12345](#) (2025) [[arxiv:2405.12278](#)]
8. *Fractionalization signatures in the dynamics of quantum spin liquids*

- K. Wang[†], **S. Feng**[†], P. Zhu, R. Chi, H. Liao, N. Trivedi, T. Xiang ([†]*Contributed Equally*)
Phys. Rev. B **111**, L100402 (2025), *Editor's Suggestion* [[arxiv:2403.12141](#)]
9. *Spin-orbit coupling controlled 2D magnetism in chromium trihalides*
 I. Lee, J. Chen, O. Molchanov, **S. Feng**, W. Huey, J. Tol, J. Goldberger, N. Trivedi, H.-Y. Kee, P. C. Hammel
Phys. Rev. B **113**, 014401 (2026) [[arxiv:2405.16709](#)]
 10. *Hidden subsystem symmetry protected states in competing topological orders*
S. Feng
Phys. Rev. B **109**, 075151 (2024) [[arxiv:2309.02307](#)]
 11. *Dimensional reduction of Kitaev spin liquid at quantum criticality*
S. Feng, A. Agarwala, N. Trivedi
Phys. Rev. Research **6**, 013298 (2024) [[arxiv:2308.08116](#)]
 12. *Machine learning reveals features of spinon Fermi surface*
 K. Zhang, **S. Feng**, Y. D. Lensky, N. Trivedi, E. A. Kim
Commun. Phys. **7**, 54 (2024) [[arxiv:2306.03143](#)]
 13. *A statistical approach to topological entanglement: Boltzmann machine representation of high-order correlation*
S. Feng, D. Kong, N. Trivedi
[arXiv:2302.03212](#) (2023)
 14. *Anyon dynamics in field-driven phases of the anisotropic Kitaev model*
S. Feng, A. Agarwala, S. Bhattacharjee, N. Trivedi
Phys. Rev. B **108**, 035149 (2023) [[arxiv:2206.12990](#)]
 15. *Detection of long-range entanglement in gapped quantum spin liquids by local measurements*
S. Feng, Y. He, N. Trivedi
Phys. Rev. A **106**, 042417 (2022) [[arxiv:2206.07237](#)]
 16. *Gapless to gapless phase transitions in quantum spin chains*
S. Feng, G. Alvarez, N. Trivedi
Phys. Rev. B **105**, 014435 (2022) [[arxiv:2012.00700](#)]
 17. *Magnetic phase transitions in quantum spin-orbital liquids*
S. Feng, N. D. Patel, P. Kim, J. H. Han, N. Trivedi
Phys. Rev. B **101**, 155112 (2020) [[arxiv:1912.09516](#)]
 18. *Film-depth-dependent Crystallinity for Light Transmission and Charge Transport in Semitransparent Organic Solar Cells*
 T. Xiao, J. Wang, S. Yang, Y. Zhu, D. Li, Z. Wang, **S. Feng**, L. Bu, X. Zhan, G. Lu
Journal of Materials Chemistry, A, **2020**, **8**, 401 (2020)
 19. *Rapidly measuring charge carrier mobility of organic semiconductor films upon a point-contact four-probes Method*
 D. Li, S. Li, W. Lu, **S. Feng**, P. Wei, Y. Hu, X. Wang, G. Lu
[IEEE J-EDS 2018.2872714](#) (2018)
 20. *Film-depth-dependent light absorption and charge transport for polymer electronics: A Case Study on Semiconductor/Insulator Blends by Plasma Etching*
 L. Bu, S. Gao, W. Wang, L. Zhou, **S. Feng**, X. Chen, D. Yu, S. Li, G. Lu
Adv. Electron. Mater **2**:1600359 (2016)

Research Experiences

Technical University of Munich

Postdoctoral Fellow

Advisors: Michael Knap, Johannes Knolle and Frank Pollmann

- Topological physics in fractal and amorphous crystals
- Disorder-free localization and many-body localization
- Dynamical signatures of fractionalization
- Quantum spin liquid candidates in rare-earth delafossite
- Machine learning assisted QMC algorithms for condensed matter theory

Garching, Germany

2024 - Present

The Ohio State University
Graduate Research Assistant and Fellow
Advisor: Nandini Trivedi

Columbus, OH, USA
2018–2024

- Theory of topological quantum matter: quantum spin liquid and Kitaev honeycomb model; detection of fractionalization; linear and non-linear response of fractionalized particles; spinon Fermi surface
- Quantum information: (topological) quantum entanglement, stabilizer code, cluster state, lattice gauge theory.
- Magnetism: quantum phase transitions and fractionalization in one dimensional frustrated systems
- Numerical methods: Exact diagonalization, matrix product states, density matrix renormalization group, time-evolving block decimation
- Statistical methods and machine learning approach to quantum many-body physics: Restricted Boltzmann machine, convolution neural network

University of California, Los Angeles
Cross-disciplinary Scholars in Science and Technology
Advisor: Prof. Hongwen Jiang

Los Angeles, CA, USA
2017-2018

- Monte Carlo simulation of electron beam induced defects in SiO₂
- Nano-imprint lithography of MOS quantum dots

University of California, Riverside
Undergraduate Research Assistant
Advisor: Prof. Marc Bockrath

Riverside, CA, USA
2016-2017

- Nano fabrication and the analysis of electronic transport in twisted bilayer graphene

Xi'an Jiaotong University
Undergraduate Research Assistant
Advisor: Prof. Guanghao Lu

Xi'an, Shaanxi, China
2016

- Transfer matrix method for light absorption in semiconductor
- In-situ reconstruction algorithm for semiconductor nano-tomography

Invited Presentations

1. *Seminar: Localization from emergent gauge theory and thermal paradox in quantum spin liquids*, Nanjing, Jiangsu, China, 2025
2. *Conference talk: Thermal paradox in quantum spin liquids*, 44th International Conference on Strongly Correlated Systems - Correlations and Coherence at Different Scales, Ustroń, Poland, 2025
3. *Seminar: From Z_2 gauge theory to quantum Majorana metal*, Wrocław University of Science and Technology, Wrocław, Poland, 2025
4. *Seminar: Dynamical features of quantum spin liquids*, Institute of Physics, Chinese Academy of Sciences, Beijing, China, 2024
5. *Lecture: Frustrated magnetism and quantum spin liquid*, Topology and Fractionalization in Magnetic Materials, Columbus, Ohio, USA, 2023

Other Conferences and Workshops

Dec. 2025: Advanced School and Conference on Quantum Matter, ICTP, Trieste, Italy

- Poster: Transient localization in a gapless spin liquid

Nov. 2025: Current Trends in Strongly Correlated and Frustrated Systems, Dresden, Germany

- Poster: Disorder-free localization of fractionalized particles

Oct. 2025: Munich Quantum Matter Days, Garching, Germany

- Poster: Transient localization by fractionalization

Sep. 2025: International Workshop on Constrained Quantum Matter, Ettal, Germany

- Poster: Anderson localization in the dynamical response of quantum magnets

Mar, 2025: DPG Spring Meeting, German Physical Society

- Contributed Talk: Emergent quantum Majorana metal from a chiral spin liquid

Jun, 2024: Gordon Research Conference: Correlated Electron Systems, Mt. Holyoke College, MA, USA

- Poster: Majorana metal from a chiral spin liquid
- Mar, 2024: APS March Meeting**, American Physical Society
- Contributed Talk: Mobility constraint of anyons in a quantum spin liquid
- Oct, 2023: Q-PHORIA**, Pittsburgh Quantum Institute, Pittsburgh, PA, USA
- Poster: Dimensional reduction of quantum spin liquids
- Jul, 2023: Boulder Summer School – Non-Equilibrium Quantum Dynamics**, CU Boulder, CO, USA
- Poster: Anyon response in field-induced quantum spin liquids
- May, 2023: TopoMag23 – Topology and Fractionalization in Magnetic Materials**, Columbus, Ohio, USA
- Poster: Anyon response in field-induced quantum spin liquids
- Apr, 2023: Topology, Symmetry and Interactions in Crystals**, KITP-UCSB, California, USA
- Poster: Dynamics of Abelian anyons in the Kitaev model
- Mar, 2023: APS March Meeting**, American Physical Society
- Contributed Talk: Transition from Kitaev quantum spin liquid to weakly coupled critical spin chains
- Feb, 2023: Edward F. Hayes Advanced Research Forum**, OSU, Ohio, USA
- Contributed Talk: Anyon, fractionalization, and their detection
- Jun, 2022: Gordon Research Conference: Correlated Electron Systems**, Mt. Holyoke College, MA, USA
- Poster: Discovery of novel topological phase in Kitaev spin liquid in a field
- Mar, 2022: APS March Meeting**, American Physical Society
- Contributed Talk: Spin response and magnetic absorption of Kitaev liquids under an external field.
- Mar, 2021: APS March Meeting**, American Physical Society
- Contributed Talk: Field-induced gapless-to-gapless phase transitions in integer spin chains.
- Aug, 2020: Ultra Quantum Matter**, Perimeter Institute for Theoretical Physics, Waterloo, Canada
- Jun, 2020: Condensed Matter Physics in all Cities**, University of Kent Canterbury, Kent, UK
- Contributed Talk: Magnetic phase transition in quantum spin orbital liquid.

Honors and Awards

- 2024: The Chinese Government Award for Outstanding Students Abroad**, Chinese Consulate in NY, USA
- The highest award granted by the Chinese government to Chinese students overseas
- 2023: Presidential Fellowship**, OSU, Columbus, OH, USA
- The Presidential Fellowship is the most prestigious award given by the Graduate School of OSU, embodying the highest standards of scholarship in the full range of Ohio State's graduate programs
- 2023: Dirac Fellowship** (offer declined), University of Florida & National MagLab, Florida, USA
- 2023: 2nd place, Edward F. Hayes Advanced Research Forum**, OSU, Columbus, OH, USA
- 2018: Siyuan Scholarship**, XJTU, Xi'an, Shaanxi, China
- Awarded to undergraduate students for their academic excellence
- 2017: CSST Scholarship**, UCLA, Los Angeles, CA, USA
- Awarded in the UCLA-CSST program for cross-disciplinary scholars in science and technology
- 2016: Meritorious Winner** of Interdisciplinary Contest in Modelling, Bedford, MA, USA
- 2016: 1st Place Award** of China Mathematical Contest in Modelling, Xi'an, Shaanxi, China

Academic Service

- Peer Reviewer**.....
- American Physical Society: Physical Review Letters; Physical Review Applied; Physical Review A; Physical Review B; Physical Review E
- IOP Publishing: Reports on Progress in Physics, New Journal of Physics

- Springer Nature Group: Nature Communication
- Others: Quantum Information & Computation

Teaching

Department of Physics, Technical University of Munich

Lecturer

Garching, Germany

Summer, 2025

- Advanced Methods in Quantum Many-Body Theory
(Topics on Lattice Gauge Theory)

Department of Physics, The Ohio State University

Graduate Teaching Assistant

Columbus, OH, USA

2018-2021

- Statistical Mechanics (Fall 2021)
- Introductory Physics – Electromagnetism, Optics, Modern Physics (Spring 2020)
- Introductory Physics – Mechanics, Thermal Physics, Waves (Fall 2019)
- Introductory Physics – Mechanics, Kinematics, Fluids, Waves (Spring 2019)
- Statistical Mechanics (Fall 2018)

Mentored Students

Department of Physics, The Ohio State University

Undergraduate Students

Columbus, OH, USA

2020-2022

- Cullen Gantenberg (Dynamics of Abelian anyons in toric code, 2022) → PhD in University of Washington
- Seth Cox (Density matrix renormalization study of spin liquids, 2022) → PhD in Boston University
- Salim Karimov (Linear response in Kitaev spin liquid, 2023) → PhD in The Ohio State University

Technical Skills

Code Library: Developer and maintainer of

- [ExactDiagPy](#): Exact diagonalization with implementation of various quantum entanglement measures

Programming Languages: Python, Julia, C++, Perl, Matlab, Mathematica, Java, Bash

Libraries and Softwares:

- Eigen, TenPy, DMRG++, ITensor, HDF5, OpenGL, Blas, Lapack; Blender, Inkscape

OS and Clusters:

- OS: Linux (Ubuntu), Windows, macOS, High Performance Computing (HPC) environments
- Clusters: Unity and Ohio Supercomputer Center (OSC)

References

Frank Pollmann
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Professor, Statistics
UCLA, USA
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